

LAPORAN UTS PEMROGRAMAN BERBASIS SERVICE

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1. Pendahuluan

Proyek ini bertujuan untuk membangun sistem reservasi lapangan olahraga menggunakan arsitektur Microservices. Fokus utama tugas ini adalah pada pembuatan API Gateway sebagai pusat kendali (Entry Point) yang menghubungkan beberapa layanan mandiri (Independent Services) melalui protokol HTTP.

2. Arsitektur Sistem

Layanan	Port	Fungsi Utama
API Gateway	5000	Routing, Proxying, dan Validasi Keamanan (JWT).
Auth Service	8000	Manajemen User, Autentikasi JWT, dan OAuth GitHub.
Field Service	8001	Pengelolaan data lapangan olahraga.
Booking Service	8002	Pengelolaan transaksi reservasi lapangan.

3. Daftar Endpoint

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Method	Endpoint	Fungsi	Akses
POST	/api/login	Login manual & dapatkan JWT	Publik
GET	/api/auth/github	Login via GitHub OAuth	Publik
GET	/api/auth/me	Ambil profil user saat ini	Private (Bearer)

Get Profile

GET ▼ http://127.0.0.1:5000/api/auth/me Send ▼

Docs Params **Authorization** Headers (14) Body • Scripts Settings Cookies

Auth Type
Bearer Token ▼

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token
Wk2Xqksyw8YkHqmM_Wf9XVCeGeg 🔗 🔒

Body Cookies Headers (9) Test Results ↺

200 OK • 577 ms • 490 B • 🌐 ⋮

{} JSON ▼ ▶ Preview 🖼️ Visualize ▼ 🔄 ☰ 🔍 📄 🔗

```
1 {
2   "id": 2,
3   "name": "Admin Test",
4   "email": "admin@test.com",
5   "email_verified_at": null,
6   "created_at": "2026-05-02T11:21:37.000000Z",
7   "updated_at": "2026-05-02T11:21:37.000000Z",
8   "github_id": null,
9   "auth_provider": "local",
10  "avatar": null
11 }
```

Refresh Token

POST ▼ http://127.0.0.1:5000/api/refresh Send ▼

Docs Params **Authorization** Headers (13) Body Scripts Settings Cookies

Auth Type
Bearer Token ▼

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token
Wk2Xqksyw8YkHqmM_Wf9XVCeGeg 🔗 🔒

Body Cookies Headers (9) Test Results ↺

200 OK • 669 ms • 875 B • 🌐 ⋮

{} JSON ▼ ▶ Preview 🖼️ Visualize ▼ 🔄 ☰ 🔍 📄 🔗

```
1 {
2   "access_token": "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJodHRwOi8vMTI3LjAuMC4xOjgwMDAvYXh0b3JlZnJlc2giLCJpYXQ1OjE3Nzc3MzIxNDM6ImV4cCI6MTc3Nzc3MzE2OCwibmJmIjoxNzc3NzMyMyJ4LCJqdGkiOiJkV3p5QnNPUXFTWEJpTXJzIiwic3ViIjoiaWIsInBydiIiOiJiZyYmQ1Yzg5NDlmNjAwYWRiMzllNzAxYzQwMDg3MmRlN2E1OTc2ZjciQkflLMjR9u_le3qoo5XmQ_6GEzDDhNp5kUTC4cpKjSK",
3   "token_type": "bearer",
4   "expires_in": 900,
5   "user": {
6     "id": 2,
7     "name": "Admin Test",
8     "email": "admin@test.com",
9     "email_verified_at": null,
10    "created_at": "2026-05-02T11:21:37.000000Z",
11    "updated_at": "2026-05-02T11:21:37.000000Z",
12    "github_id": null,
13    "auth_provider": "local",
14    "avatar": null
15  }
16 }
```

Logout

POST ▼ http://127.0.0.1:5000/api/logout Send ▼

Docs Params **Authorization** Headers (14) Body • Scripts Settings Cookies

Auth Type
Bearer Token ▼

Token
tNnQgYNs-z4s6A7Xhp2tux0Sg0W3Q 🔗 🔒

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (9) Test Results ↺ 200 OK • 644 ms • 329 B • 🌐 ⋮

{} JSON ▼ ▶ Preview 🖼 Visualize ▼ ↺ ≡ 🔍 📄 🔗

```
1  {}
2  {"message": "Berhasil logout (Token dihapus dari session)"}
3  {}
```

GET fields (halaman 1)

GET ▼ http://127.0.0.1:5000/api/fields Send ▼

Docs Params **Authorization** Headers (14) Body • Scripts Settings Cookies

Auth Type
Bearer Token ▼

Token
tNnQgYNs-z4s6A7Xhp2tux0Sg0W3Q 🔗 🔒

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (9) Test Results ↺ 200 OK • 717 ms • 5.64 KB • 🌐 ⋮

{} JSON ▼ ▶ Preview 🖼 Visualize ▼ ↺ ≡ 🔍 📄 🔗

```
272  },
273  {
274    "url": "http://127.0.0.1:8001/api/fields?page=2",
275    "label": "2",
276    "page": 2,
277    "active": false
278  },
279  {
280    "url": "http://127.0.0.1:8001/api/fields?page=2",
281    "label": "Next &raquo;",
282    "page": 2,
283    "active": false
284  }
285 ],
286 "next_page_url": "http://127.0.0.1:8001/api/fields?page=2",
287 "path": "http://127.0.0.1:8001/api/fields",
288 "per_page": 10,
289 "prev_page_url": null,
290 "to": 10,
291 "total": 12
292 }
```

GET fields (halaman 2)

GET <http://127.0.0.1:5000/api/fields?page=2> Send

Auth Type: Bearer Token

Token: `tNnQgYNs-z4s6A7Xhp2tux0Sg0W3Q`

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body: `{ "current_page": 2, "data": [{ "id": 11, "category_id": 3, "name": "Bulutangkis Jaya Bersama", "type": "Badminton", "price_per_hour": 40000, "created_at": "2026-05-02T21:23:16.000000Z", "updated_at": "2026-05-02T21:23:16.000000Z", "category": { "id": 3, "name": "Badminton", "created_at": "2026-05-02T21:10:13.000000Z",`

GET fields dengan filter

GET http://127.0.0.1:5000/api/fields?category_id=2 Send

Auth Type: Bearer Token

Token: `tNnQgYNs-z4s6A7Xhp2tux0Sg0W3Q`

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body: `{ "url": "http://127.0.0.1:8001/api/fields?page=1", "label": "1", "page": 1, "active": true }, { "url": null, "label": "Next »", "page": null, "active": false }, ], "next_page_url": null, "path": "http://127.0.0.1:8001/api/fields", "per_page": 10, "prev_page_url": null, "to": 4, "total": 4 }`

Get detail fields

GET <http://127.0.0.1:5000/api/fields/1> Send

Auth Type: Bearer Token

Token: `tNnQgYNS-z4s8A7Xhp2tux0Sg0W3Q`

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body: `{}` JSON Preview Visualize

```
1 {
2   "id": 1,
3   "category_id": 1,
4   "name": "Jakarta International Stadium",
5   "type": "Football",
6   "price_per_hour": 750000,
7   "created_at": "2026-05-02T21:23:16.000000Z",
8   "updated_at": "2026-05-02T21:23:16.000000Z",
9   "category": {
10    "id": 1,
11    "name": "Football",
12    "created_at": "2026-05-02T21:10:13.000000Z",
13    "updated_at": "2026-05-02T21:10:13.000000Z"
14  },
15  "facilities": [
```

200 OK • 574 ms • 1.24 KB

Tes Github OAuth 2.0

Authorize UTS PPLOS - Taraka Zubair Gama

UTS PPLOS - Taraka Zubair Gama by [tarakagama](#) wants to access your tarakagama account

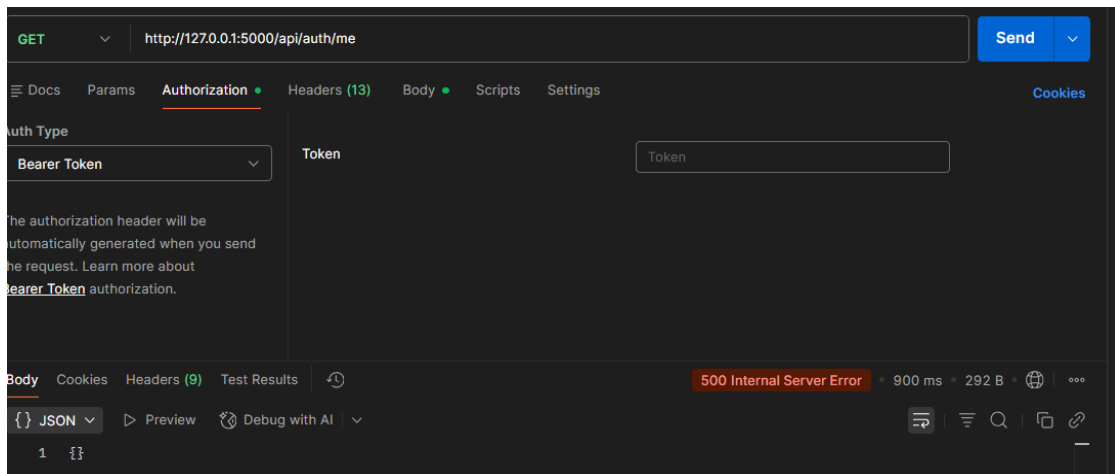
Personal user data ▼
Email addresses (read-only)

Cancel Authorize tarakagama

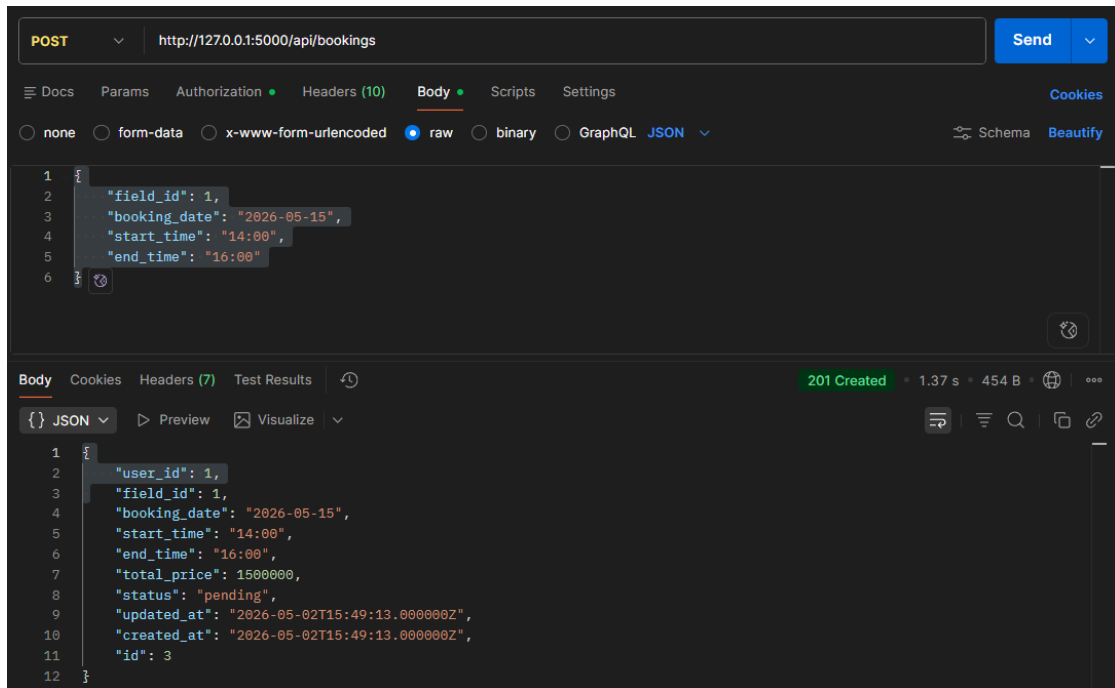
Authorizing will redirect to <http://127.0.0.1:8000>

Not owned or operated by GitHub Created 4 days ago Fewer than 10 GitHub users

Get tanpa bearer token



Post Booking



4. Kesimpulan

Berdasarkan hasil perancangan, implementasi, dan pengujian sistem Field Service Microservices, dapat disimpulkan beberapa poin utama sebagai berikut:

- **Arsitektur Microservices Berhasil Diimplementasikan:** Sistem telah berhasil dipecah menjadi beberapa layanan mandiri (*Auth-Service*, *Field-Service*, *Booking-Service*) yang berkomunikasi melalui API Gateway sebagai satu-satunya pintu masuk utama (*Single Entry Point*) pada port 5000.

- Keamanan Terpusat dengan JWT: Implementasi autentikasi menggunakan *JSON Web Token* (JWT) memastikan setiap pertukaran data antar-layanan aman. Fitur *Login*, *Logout*, *Refresh Token*, dan *Get Profile* telah berhasil melewati proses *proxying* dari Gateway ke Auth-Service dengan validasi token yang akurat.
- Efisiensi Pengelolaan Data: Penggunaan relasi database yang kompleks (melibatkan tabel *Categories*, *Fields*, *Facilities*, dan *Schedules*) memungkinkan sistem menyajikan informasi lapangan secara komprehensif. Fitur *Pagination* dan *Filtering* yang diimplementasikan pada *Field-Service* terbukti mampu mengoptimalkan performa pengiriman data JSON pada sisi klien.
- Keamanan Gateway (Rate Limiting): Fitur *Rate Limiting* (*Throttling*) pada API Gateway berhasil melindungi sistem dari serangan *brute force* atau *spamming request*, dengan membatasi jumlah permintaan maksimal 60 kali per menit untuk setiap pengguna.
- Standarisasi Pengembangan: Seluruh proses pengembangan dilakukan dengan standar industri, mulai dari penggunaan *Conventional Commits* pada Git, pengelolaan versi menggunakan *Git Tagging*, hingga pendokumentasian API yang terstruktur.